

Optimal Wiring of Heliostats in Solar Tower Power Plants

Lab-course Computer Science
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Motivation



Figure: Solar tower power plant. Heliostats reflect onto a receiver in order to heat up a medium for energy conversion.

Sun's position changes continuously, thus for effective energy conversion we need to rotate the heliostats.

⇒ each heliostat requires a data and power cable

Problem

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- ▶ Goal: Find subgraphs $H_{\text{data}}, H_{\text{power}} \subseteq G$ with minimal total costs that both connect all heliostats with the solar tower.
 - ▶ data cable: cable and switch costs
 - ▶ power cable: cable costs

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 - ▶ data cable: cable and switch costs
 - ▶ power cable: cable costs
- ▶ Constraints:
 - ▶ no intersecting edges
 - ▶ cables have capacity and length constraints
 - ▶ switches limit connectivity at heliostats for data cables

Problem: Data cable

Table: Data cables with their limitations and costs. We can only connect two heliostats with a copper cable.

Cable	Max. capacity	Max. length [m]	Price [€/m]
Copper	2	100	0.7
Fiberglass	-	-	2

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Table: Switch prices, for 8-/16-port switches we can either use copper or fiber-glass output. Endpoints are only used for copper cables.

Switch	Max. degree	Price [€/p. piece]
Endpoint	1	10
Conductor	2	100
8-port copper/fiberglass switch	9	800
16-port copper/fiberglass switch	17	1500

Problem: Power cable

Table: Power cables with their connectivity limitations and prices.

Cable	Max. capacity	Max. length [m]	Price [€/m]
NY-Y-J 3x2.5 RE	56	38.36	0.58
NY-Y-J 3x4 RE	73	47.08	0.87
NY-Y-J 3x6 RE	92	56.04	1.24
NY-Y-J 3x10 RE	124	69.30	1.95
NY-Y-J 3x16 RE	162	84.87	3.13
NY-Y-J 3x25 RM	209	102.79	5.19
NY-Y-J 3x35 RM	250	120.31	6.90

- For instances with more than 250 heliostats we need to use more than one outgoing cable from the solar tower
⇒ Partitioning problem

Problem: Installation cost

Table: Depending on the location we have different installation costs for trenches and working power.

Country	Price [€/m]
Australia	50
Spain	25
South Africa (RSA)	10
United Arab Emirates (UAE)	10

► additionally: protective foil (+2€/m)

Thus for a meter of glasfiber in Australia we need to pay

50€ installation cost

+2€ glasfiber cable

+2€ protective foil

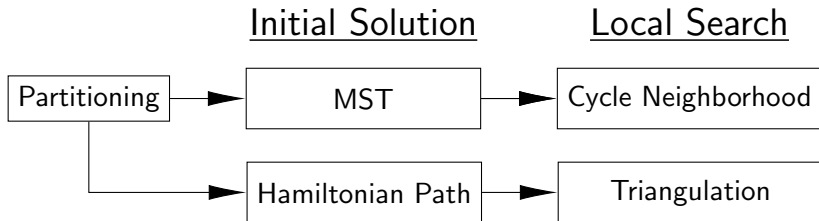
54€

Our approach

- ▶ Consider a precomputed partitioning $\Rightarrow$ eases capacity constraint and computation time
- ▶ Focus on data cabling due to high switch costs
 - ▶ compute solution only on data cable
 - ▶ afterwards compute power cable assignment greedily with data cable solution
 - ▶ omit copper cables for simplicity (can be added afterwards) in order to save a bit of money

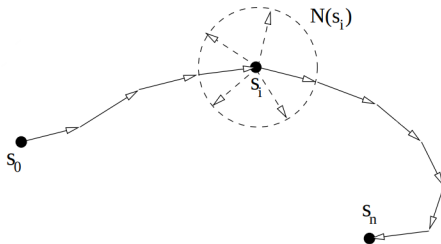
$\Rightarrow$ consider solution that installs data and power cable together in trenches, that is, we only have one subgraph $H \subseteq G$

Our approach



► Local search

- search space = feasible solutions to an optimization problem
- from solution s_i jump to promising solution within neighborhood $N(s_i)$.



Partitioning

Maximum number of connected heliostats per power cable is limited by = 250 (max. capacity).

- ▶ Partition heliostats beforehand: Calculate angles between heliostat-vectors and the x-axis.
- ▶ Compute initial solutions on each partition independently.

Partitioning

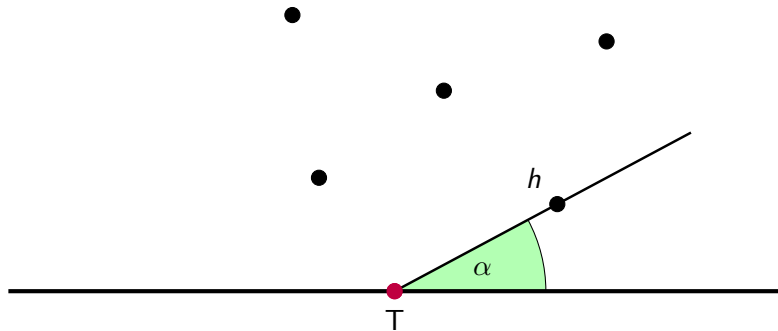


Figure: Compute angle between x-axis and $\text{line}(T, h)$, $h \in V_H$.

Partitioning

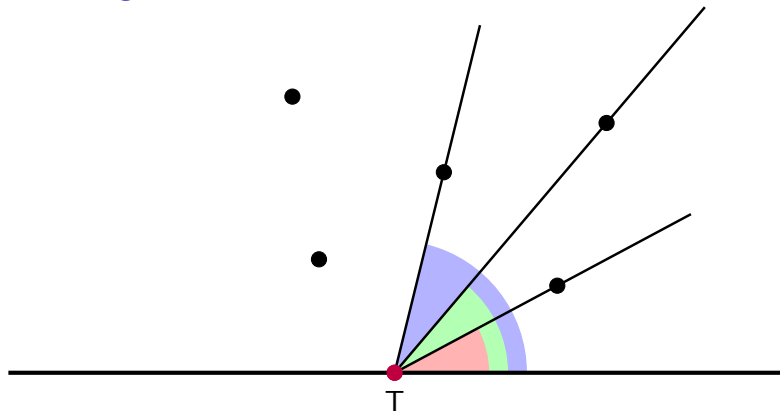


Figure: Sort heliostats by computed angles. For $k \in \mathbb{N}$ partitions we obtain $\left\lceil \frac{|V_H|}{k} \right\rceil$ heliostats in each partition.

Heuristic: Minimum Spanning Tree (MST)

Initial Solution: MST

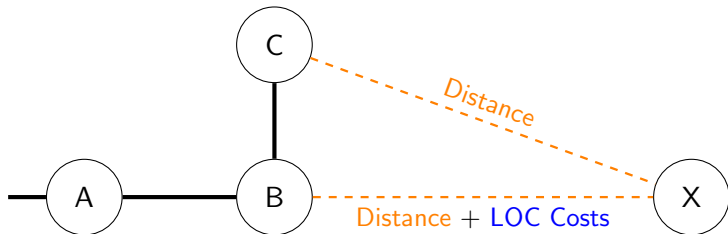


Figure: Build MST with Prim's algorithm. Include switch installation into edge costs. Note: We only consider glasfiber cables for simplicity.

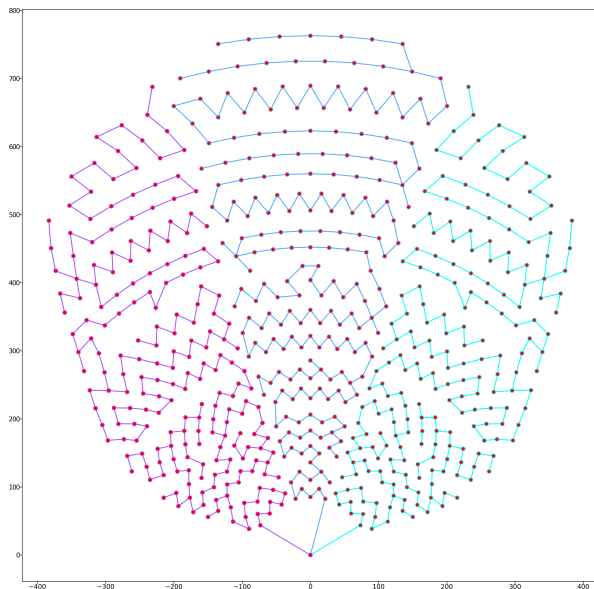


Figure: Initial MST solution for PS10 instance with 3 partitions and 54€/m edge costs. Total costs: 981,231.36 € (data cables and switch costs).

Local Search: Cycle Neighborhood

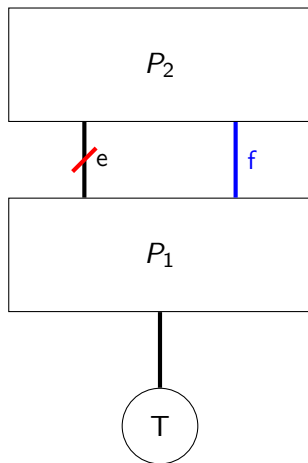


Figure: Consider the cycle neighborhood, that is, delete edge e and reconnect the resulting partitions P_1 and P_2 by adding an edge f with $c(f) < c(e)$.

Local Search: Cycle Neighborhood - Power Cables

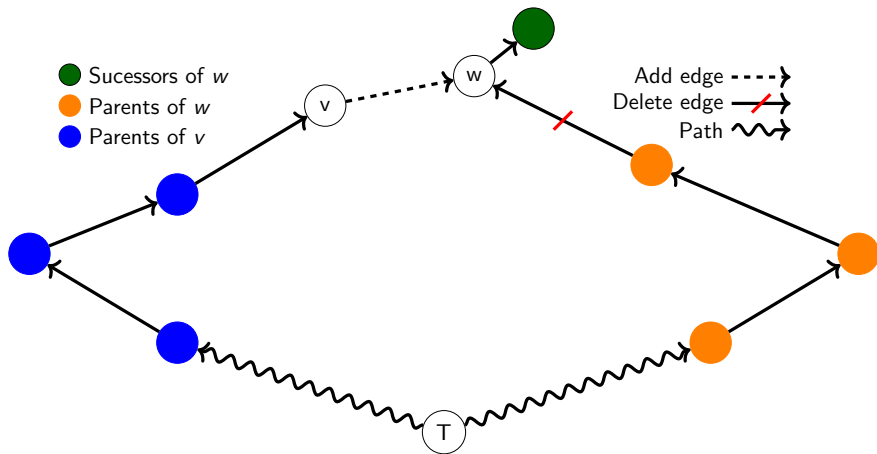
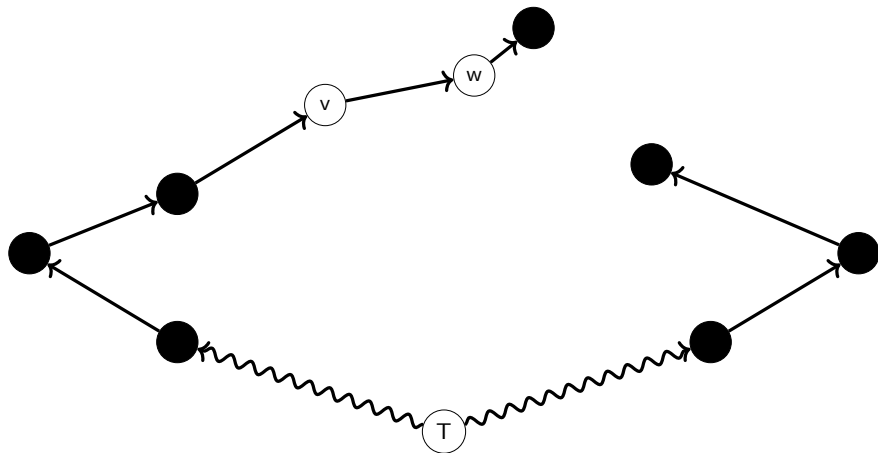
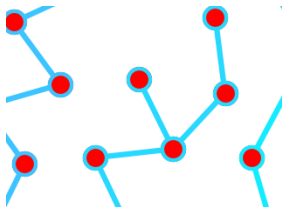


Figure: For power cables we need to consider up- and downgrades regarding cables $\Rightarrow$ potential savings by switching to lower-quality cables.

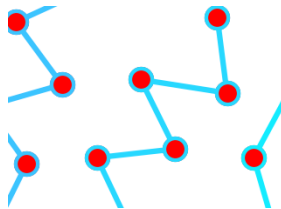
Local Search: Cycle Neighborhood - Power Cables



Cycle Neighborhood Improvement



(a) Initial solution.



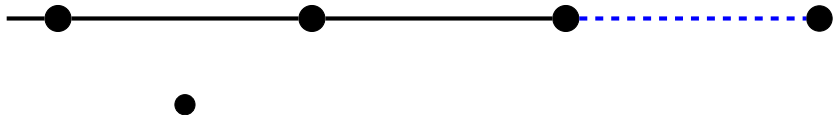
(b) After local search improvement. A switch is replaced by a Conductor.

Heuristic: Hamiltonian Path

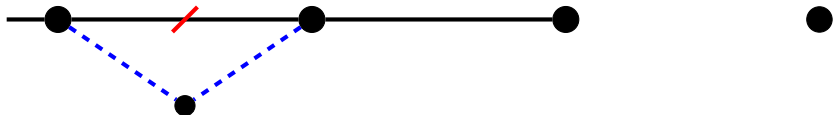
Initial Solution: Hamiltonian Path

i) Append

add edge - - -



ii) Insert



Local Search: Hamiltonian Path – Triangulation

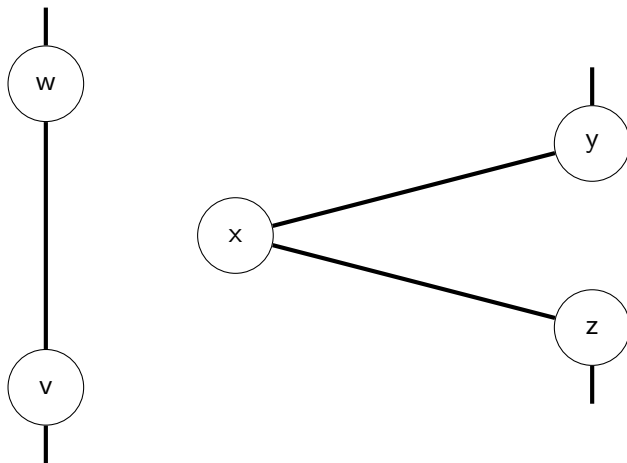


Figure: Starting point.

Local Search: Hamiltonian Path – Triangulation

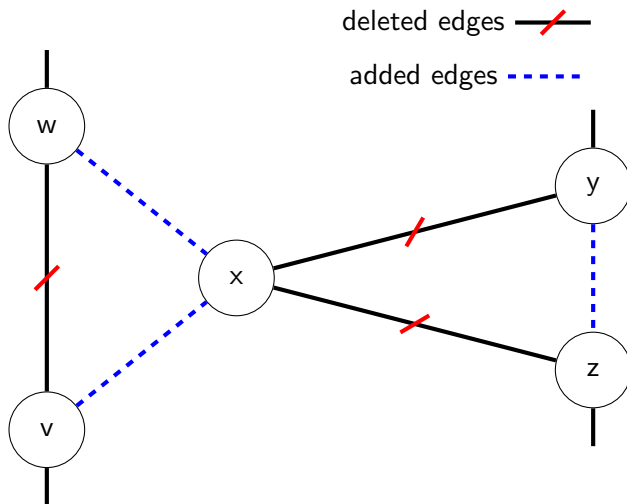


Figure: Check costs of added and deleted edges.

Local Search: Hamiltonian Path – Triangulation

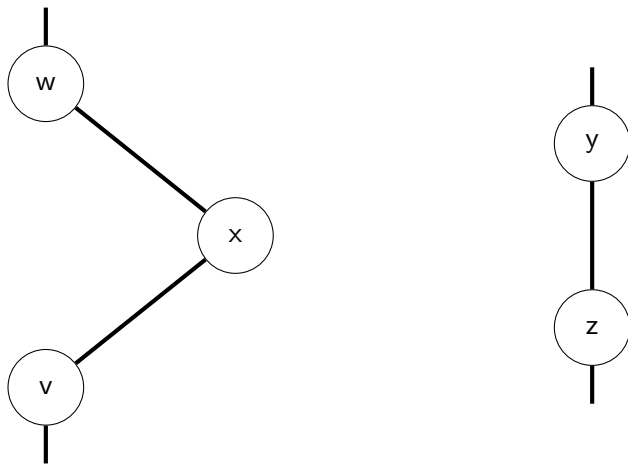
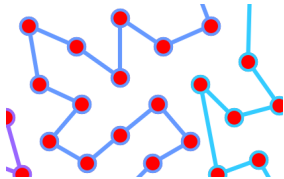
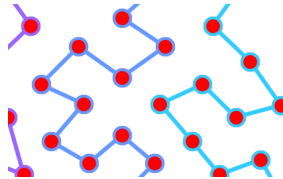


Figure: Added edges have lower costs than deleted ones.

Triangulation Neighborhood Improvement



(a) Initial solution.



(b) After triangulation improvement.

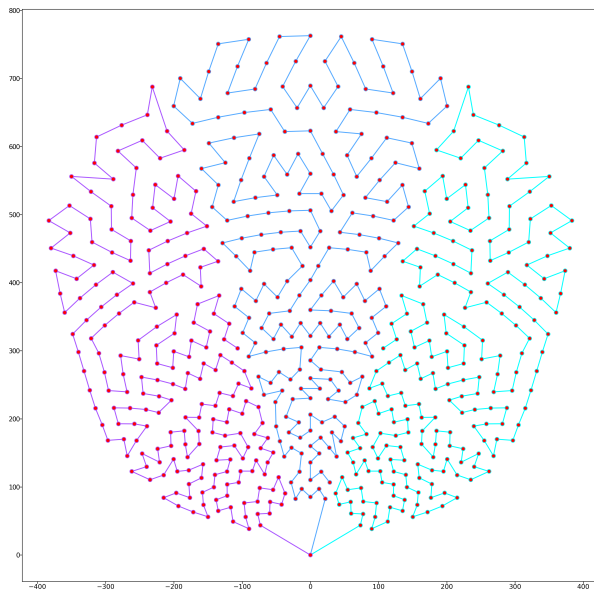


Figure: Initial Hamiltonian paths solution for 3 partitions and 54€/m.
Total costs: 992,324.18 € (data cables and switch costs).

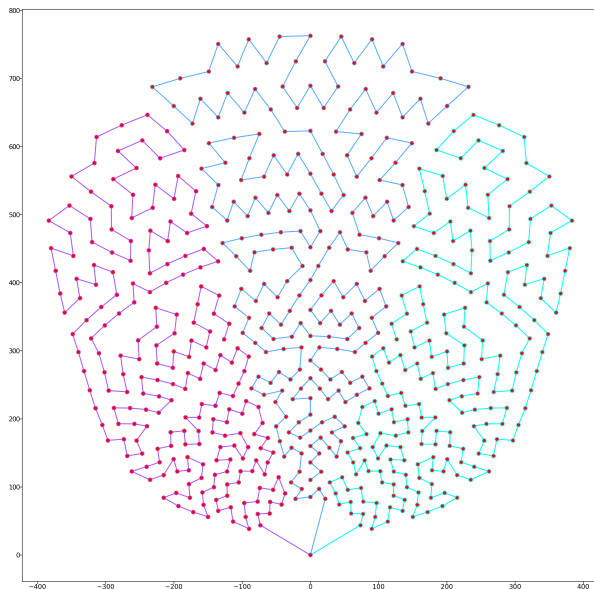


Figure: After local search for Hamiltonian paths. Total costs: 971,016.47€ (data cables and switch costs).

Evaluations

Comparison for 3 Partitions after Local Search: MST

Table: Total costs for 3 partitions. We boldfaced the best initial values for each country.

Algorithm	Australia cost[€]	Spain cost[€]	UAE/RSA cost[€]
Hamiltonian Path	993,745.675	573,402.55	320,853.59
MST	1,021,309.88	611,592.39	362,567.34

Table: Improvement of total costs by Local Search algorithm with 3 partitions.

Algorithm	Australia	Spain	UAE/RSA
Hamiltonian Path	2.1%	1.9%	1.7%
MST	1.0%	1.4%	1.4%

Australia

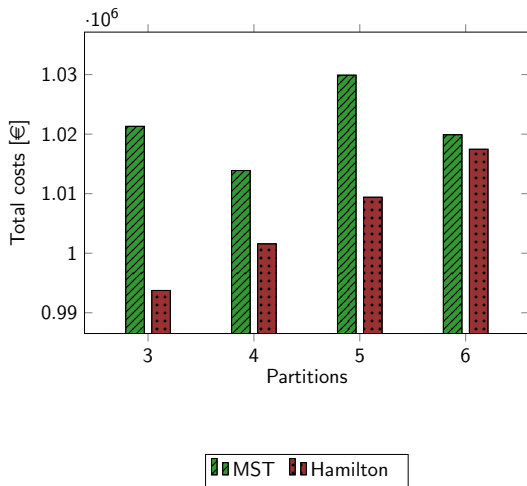


Figure: Computed on PS10 instance. Installation costs: 50€/m.

Spain

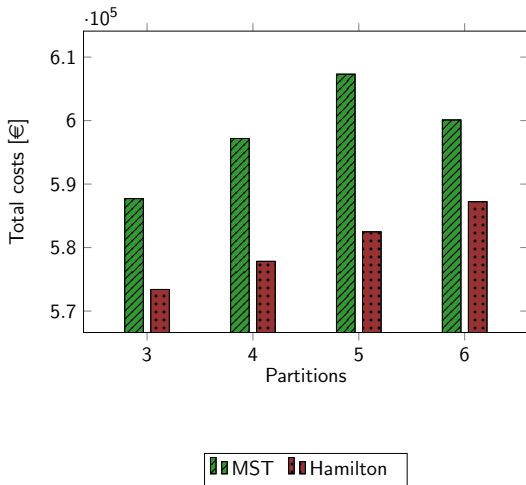


Figure: Computed on PS10 instance. Installation costs: 25€/m.

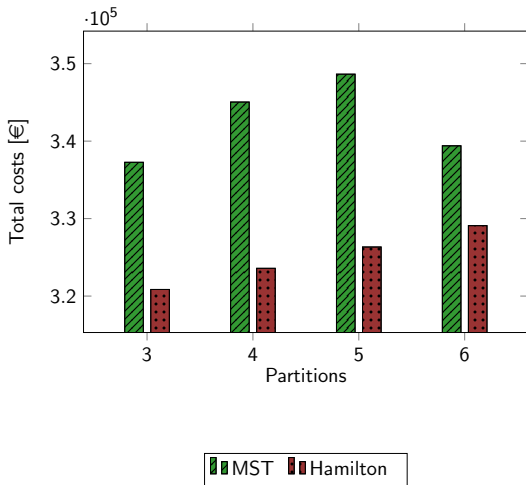


Figure: Computed on PS10 instance. Installation costs: 10€/m.

Conclusion

- ▶ heuristics yield fast, valid solutions as opposed to its mixed-integer program
- ▶ MST local search procedure too restrictive
 - ▶ have to consider up- and downgrades of power cables
 - ▶ switch costs eat up a lot of budget!
- ▶ Hamilton local search → greedy power cable assignment yield best results
- ▶ considering both as separate problems might be also beneficial
 - ▶ local search is not too restrictive
 - ▶ power cabling problem $\equiv$ multilevel capacitated minimum spanning tree¹

¹Gamvros et al.: The multilevel capacitated minimum spanning tree problem, INFORMS Journal on Computing, 2006

Thank you for your attention!

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